FASTER MODELS
The Bell X-1A and X-1B were improved models of the experimental rocket aircraft that were produced between 1949 and 1952. They could carry more fuel than the original X-1, and the redesigned cockpit canopy gave the pilot a better view. Yeager reached Mach 2.44 in the X-1A in December 1953.

paving stones." At .94 Mach, the pitch control went dead as the shockwave immobilized his elevators – a solution was found in controlling pitch through tilting the stabilizer, the horizontal part of the tail assembly. At .96 Mach, the project came close to disaster in an incident unrelated to speed, when Yeager's cockpit windshield iced over at 43,000ft (13,000m), completely blocking his view; he had to make a blind landing with wholly inadequate instruments, talked down by the pilot of the chase aircraft shadowing his flight.

Finally broken

At last, on October 14, 1947, the famous sound barrier was passed without incident – so easily that Yeager recorded a feeling of letdown. "There should've been a bump on the road," he wrote, "something to let you know you had just punched a hole through that sonic barrier." But the only way he knew that he had reached Mach 1 was because his cockpit instruments, and those of the NACA engineers monitoring the flight, said so.

Breaking the sound barrier was only a start. Fliers at Edwards Air Force Base went on pushing speed and altitude records to the limit. In the early 1950s the Douglas D-558-2 Skyrocket began to set the pace, flown by US Navy pilot Bill Bridgeman and NACA test pilot Scott Crossfield. With a small swept wing and almost perfect streamlined shape, the Skyrocket reached Mach 1.88 – 1,238mph (1,992kph) at 66,000ft (20,100m) – in August 1951, and passed the new milestone of Mach 2 in November 1953. Fired by the competitive spirit so crucial to the "right stuff," Yeager quickly struck back with the Bell X-1A, recording a speed of Mach 2.44 the following December.

Perilous pursuit

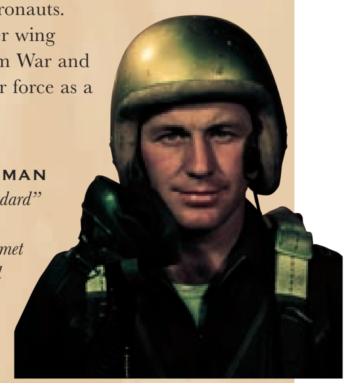
These rocket-plane flights, pushing back the limits of aviation technology, continued to be extremely hazardous. The Skyrocket suffered from "supersonic yaw," which would suddenly send the plane skidding on an oblique course. The Bell X-1s were plagued by mystery explosions, which were eventually traced to a minor but fatal flaw in gaskets in the rocket system. But this was only one of, in Yeager's words, "a dozen different ways that the X-1 can kill you." Yeager himself survived total loss of control in the X-1A during his record-setting Mach 2.44 flight of December 1953 – pushed too hard, the aircraft tumbled 50,000ft (15,250m) before he was able to wrestle it back under control. Another test pilot, Milburn Apt, was less fortunate under similar circumstances. In September 1956 Apt took the Bell X-2 to Mach 3.2, becoming the first person to fly at over 2,000mph (3,218kph). But setting the record killed him. The X-2 plummeted from high altitude out of control and, slipping in and out of consciousness, Apt was unable to bail out before hitting the ground.

CHUCK YEAGER

Charles Elwood "Chuck" Yeager (1923–) was born in 1923 in the backwoods town of Hamlin on the Mud River, West Virginia. In 1941, fresh out of high school, he joined the air force and trained to be a fighter pilot. Sent to Europe in 1943, Yeager was shot down over Nazi-occupied France but escaped back to England via neutral Spain. He returned to combat during the invasion of Normandy. Flying Mustangs, he claimed 13 and a half kills, five of them in one day, including among his victims a Messerschmitt Me 262 jet fighter.

After the war Yeager trained as a test pilot at Wright Field in Dayton, Ohio. Selected to fly in the X-1 program at Muroc, he became the first man to break the sound barrier in October 1947. When his achievement was belatedly made public, he handled the fame and honors with the same quiet laid-back confidence that he had brought to piloting a rocket plane. He set a further speed record in 1953, reaching Mach 2.44 in the X-1A, but was also content to take the support role, often flying "chase" in a fighter while another pilot flew the test aircraft.

Yeager was unusal among test pilots in not having a college education. This lack of formal qualification prevented him from being chosen as one of the first astronauts. Yeager led a fighter wing during the Vietnam War and retired from the air force as a general in 1975.



HELMETED AIRMAN
Shown here in a "standard" pilot's helmet, Yeager improvised his first helmet using a leather football helmet. He cut holes in it to accommodate his earphones and oxygen supply.